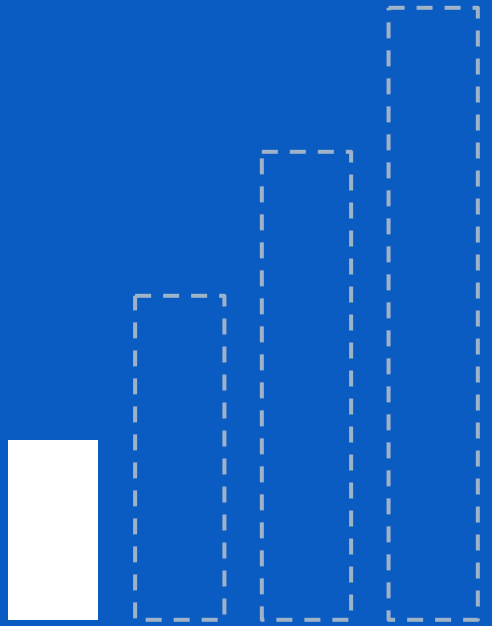
UNSTRUCTURED DATA // FINAL PROJECT // OCT 15, 2026

TRIAL SIGNAL

Are rare-disease clinical trials measuring what patients say matters most?

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1 IN 4

patient priorities the average trial measures

Do trials measure what patients care about?

Endpoints are what a trial measures

They decide what a new drug can claim to improve once it is approved.

The FDA asks patients what matters

Patients describe their symptoms and what they want treatment to fix. The FDA publishes this as a report.

Comparing the two by hand is too slow

The reports are long and endpoints are free text, so we used NLP to compare them.

PATIENT PRIORITY

“Ability to walk independently”

TRIAL ENDPOINT

“Change from baseline in 6-Minute Walk Test at Week 48”

Different words, same goal. Our model compares meaning, not keywords.

OUR QUESTION: For each trial, how many of the things patients care about does it actually measure?

We used two sources of text data

SOURCE 01

FDA Voice of the Patient reports

Seven long reports, one per disease, where patients describe their symptoms, daily life and hopes for treatment.

102 patient priorities found

73 73 selected for endpoint-alignment analysis

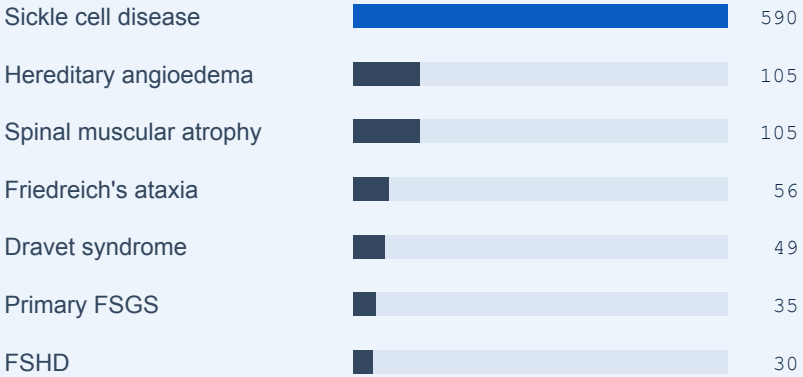
29 kept as background

Some priorities, like travel burden, matter to patients but aren't something a trial measures. We kept them as context but left them out of the score.

SOURCE 02

ClinicalTrials.gov (API v2)

970 trials **8,445** endpoints



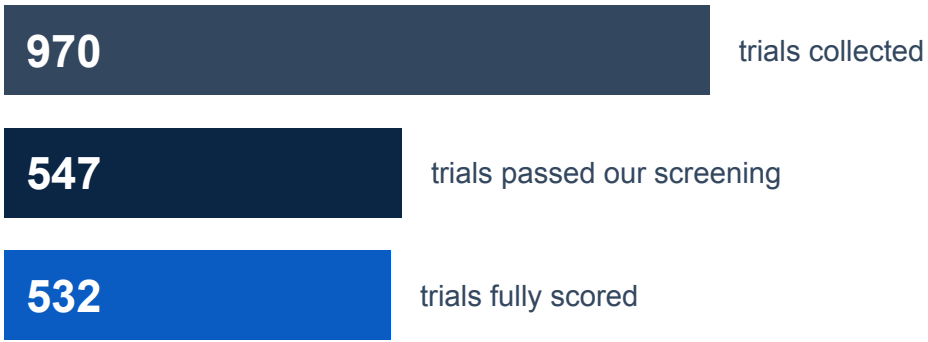
We cleaned both sources before comparing them

PATIENT REPORTS

PDF → Chunk → Embed → RAG retrieve → LLM extract → Human review → **73 priorities**

For every priority we kept the exact quote and page number, so anyone can check it against what patients said.

TRIALS



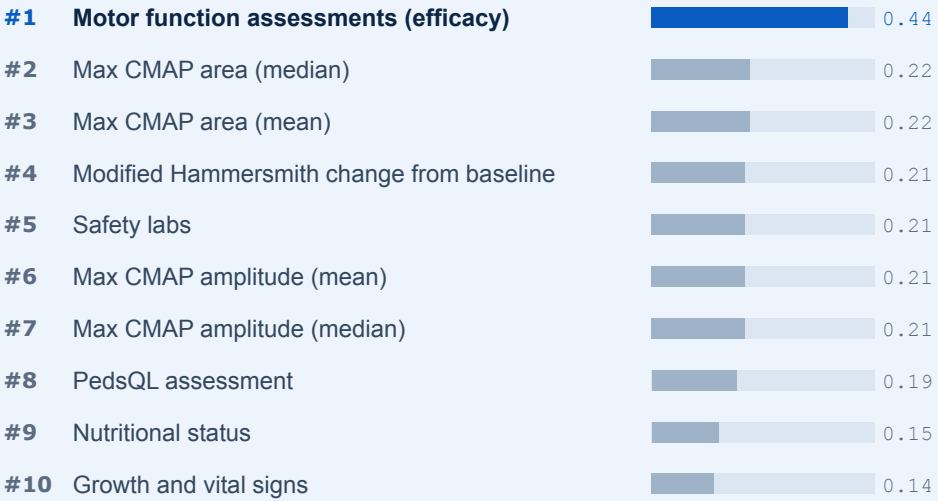
WHAT WE TOOK OUT

- Trials for the wrong disease that showed up in our search
- Trials that mix several patient groups
- Trials that were withdrawn or enrolled no one
- Studies that test diagnosis or biology, not a treatment
- 15 trials had unclear results, so we left them out instead of scoring them zero

Step 1: use NLP to shortlist plausible endpoints

MiniLM embeddings + cosine similarity rank each trial's endpoints for one patient priority. Keep up to 10; no paid LLM here.

EXAMPLE **"Preserve mobility"** in an SMA trial



15/16

real matches were ranked #1

16/16

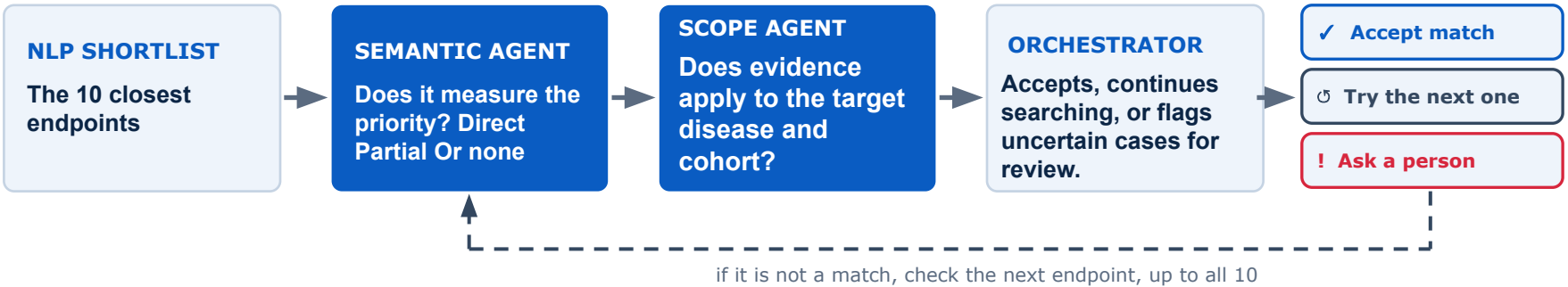
confirmed direct matches in the top 10

Endpoint-count bias substantially reduced

More endpoints no longer means a higher score

Cosine retrieves candidates; it is NOT the final alignment decision.

Step 2: Agent pipeline verifies and scores candidate matches



SCORING AGENT = DETERMINISTIC PYTHON

Trial score = **(direct + ½ partial)** ÷ priorities

Example: out of 9 priorities, a trial fully covers 1 and partly covers 3. Score = (1 + 1.5) ÷ 9 = 28%.

~37,000 AI checks over 2 batch runsl

0 / 38 non-matches wrongly called a match in our test

5,794 priority-and-trial pairs judged

[06] RESULTS

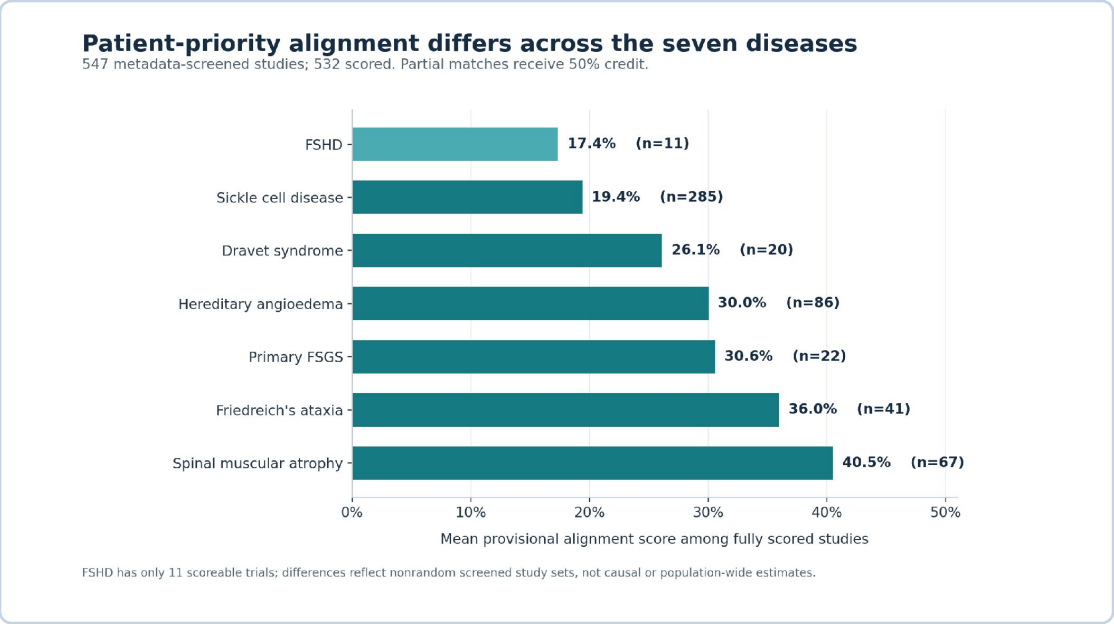
Clinical trials show only 25.7% average alignment with patient priorities.

25.7%

On average, a trial in our sample measures about a quarter of what its patients said matters (532 trials).

65% of all checks found no registered endpoint that matched

52 trials had no registered endpoint matching any patient priority



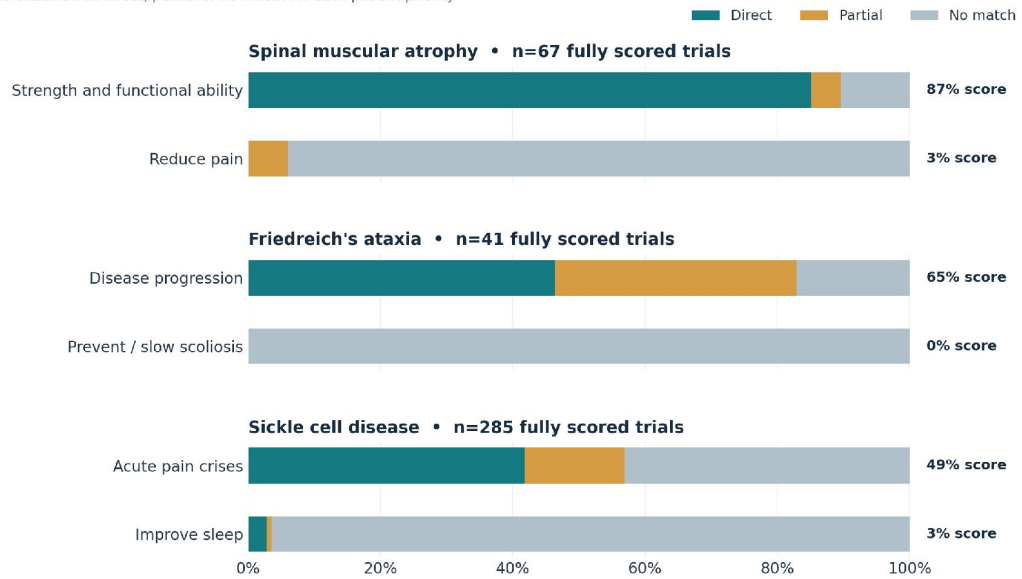
Early results: we score what trials say they measure, not whether drugs work. Counting each disease equally, the average is 28.6%.

Trials prioritize disease outcomes over everyday symptoms.

Within the same disease, trials measure one patient priority often and another almost never.

The same disease can measure one patient priority and overlook another

Share of trials classified as direct, partial or no match for each patient priority



Labels are based on registered endpoint text and the frozen V5/Scope pipeline; 50%-credit scores shown at right.

BY TYPE OF PRIORITY

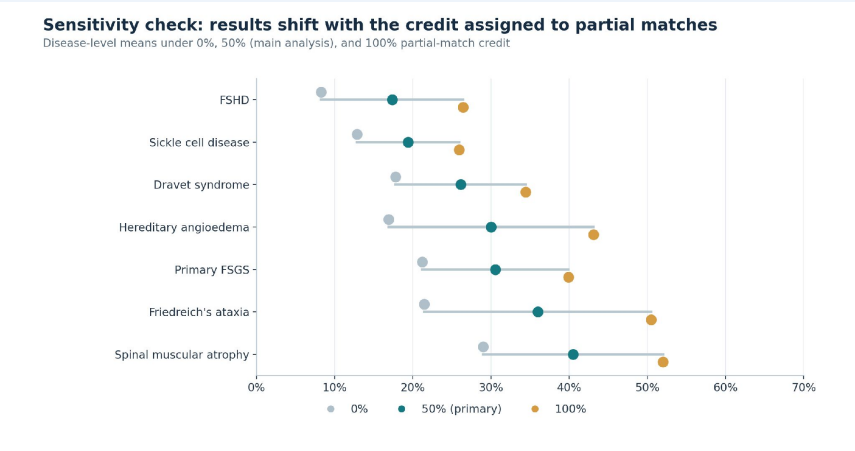
Disease control	47%
Daily function	37%
Symptoms	17%
Emotional	9%
Thinking & memory	5%

FATIGUE

Measured in 12% of trials or fewer, in all 5 diseases where patients mentioned it.

Does the result hold up?

AVERAGE SCORE, BY HOW MUCH A PARTIAL MATCH COUNTS



The conclusion holds

However we count partial matches, trials measure well under half of what patients care about.

Pooled means: 16.6%–34.8%

285 of 532 trials, so we also show every disease on its own.

Unclear cases aren't counted as zero

15 trials with unresolved results are left out of the averages.

Not yet reviewed by doctors

Trial screening used registry information, not clinical review.

The math checks out

We re-calculated every score and all 532 match exactly.

Who can use this, and how?

Drug companies
Check a planned trial before it starts, and add a measure for what patients care about but is missing.

FDA reviewers
Quickly compare a trial's endpoints with what patients told the FDA about that disease.

Patient groups
Back up their requests with data. For example, 63 of 67 SMA trials don't measure pain.

IDEA: TRIAL DESIGN CHECKER

Example: checking a new SMA trial

✓ Strength & functional ability	87%
✓ Preserve mobility	83%
✓ Respiratory function	37%
✗ Fatigue & energy	10%
✗ Contractures & scoliosis	6%
✗ Pain	3%
✗ Communication	3%

% = score across 67 past SMA trials (full match = 1, partial = ½)

KEEP IN MIND We measure what trials say they track, not whether the drug works. Results are early and still need review by doctors.